

- Step 4. Knowing the SIL concentration, noted X_{SIL} or equivalently X_c , used for spiking the working sample, direct quantification of the analyte Z can be provided without the need for a calibration curve as shown in Equation (1.7).

Estimated sample concentration

$$Z = X_{SIL} \times \frac{q_{nat}}{q_{lab}} \quad (1.7)$$

To correctly achieve this procedure and be able to perform the deconvolution, it is essential to have a crosstalk or isotopic overlap. This is possible when SIL chemical purity and/or isotopic enrichment is less than 100% or when there is only a small mass-unit difference between the isotope labeled standard and its analogous compound. IPD reproducibility was estimated based on an interlaboratory study, including four different World Anti-Doping Agency (WADA) accredited laboratories, and compared to a more traditional EC calibration method using surrogate standards. More details on interlaboratory precision parameters are available in Section 3.1.

The IPD shows the same accuracy and demonstrates improved reproducibility at low concentrations (2 ng/ml) with a relative standard deviation of reproducibility ranging approximately from 10 to 16%, respectively [16]. This result shows that isotope dilution mass spectrometry determination analytical methods are of high metrological quality. To confirm the high metrological reliability of the IPD, MU was estimated the same manner it is presented in Section 6.4 for LEAD example.

Uncertainty budget shows that MU is mainly dependent on the experimental determination of isotopic abundance (78.0%) and SIL concentration measurement (21.3%). Reducing these two sources of uncertainty involves some additional work during method development, but the IPD procedure then benefits from a better performing and faster analysis because calibration is performed within the working sample, and no traditional EC curve is required.

1.5.2 Direct Internal Calibration with Labeled Calibrant (IC-SIL)

When possible, the simplest quantification procedure is probably achieved when an in-sample single amount of surrogate calibrant is used to compute the working sample concentration. With this procedure, authentic and surrogate standards are simultaneously measured. The estimated analyte concentration, Z , is directly obtained via the peak area ratio of the sample *versus* the surrogate calibrant. Because only one concentration level is introduced in the working sample, a response factor relationship must be first established to confirm the absence of ionization competition between surrogate and authentic analyte, independent of the concentration. Thus, equimolar mixtures of surrogate and authentic analyte in neat, artificial and/or depleted matrices are first analyzed over the investigated calibration range. Additionally, ionization competition at nonequimolar concentrations should be investigated. Thus, several multipoint calibrations using the authentic analyte with surrogate standard at different concentration levels can be analyzed to study the authentic analyte response function alteration. Once the RF has been empirically determined, the working sample concentration is calculated as follows:

Working sample concentration

$$Z = \frac{Y}{Y_{SIL}} \times \frac{X_{SIL}}{RF} \quad (1.8)$$

This equation is a reorganization of Equation (1.3), where the intercept is zero, and the slope a_1 corresponds to the RF . When SIL is spiked at low concentration, such as 12.5 or 25% of the ULOQ, marked competitive ion suppression occurs due to the concomitant presence of the analyte at higher concentrations in ionization source. Conversely, when the SIL concentration is fixed in the highest bound of the response function, the influence of the surrogate signal on a low concentrated analyte can be detrimental and generate a significant bias. A correction procedure was proposed by determining the SIL concentration equivalent, noted X_{SIL}^* obtained with the following formula:

SIL concentration equivalent

$$X_{SIL}^* = X_{SIL} \times P_{SIL} \times E_{SIL} \times \frac{MW_A}{MW_{SIL}}$$

where MW is the molecular mass of authentic analyte and SIL surrogate, P the chemical purity as percentage, and E the isotopic enrichment, expressed as the probability of finding a labeled atom at any single site [17]. New reagents and improved instrumentation give opportunities to develop novel and faster quantification procedures exhibiting high metrological quality parameters. For instance, the one-point calibration method using SIL as calibrant and their isotopes was introduced to extend the lower limit of quantification (LLOQ).

To perform this analysis, a triple quadrupole instrument was used and a particular acquisition method named multiple isotopologue reaction monitoring (MIRM) was developed. By monitoring the SIL isotopic fragmentation abundances, a regression model was constructed by plotting the surrogate standard concentration equivalent on the abscissa and the instrument response (peak areas) of the corresponding MIRM channel on the ordinate. Then, the authentic analyte concentration can be calculated using the regression parameters [18]. This is just an example of regularly active literature.

Overall, internal calibration with SIL as calibrant is conceptually straightforward for absolute quantification with modern MS instrumentation, but requires additional steps during method development, such as the experimental determination of the RF and, with the MIRM procedure, isotopic abundance determination. However, once the method is developed, it is markedly faster in routine analysis because a daily repeated calibration curve is no longer required, and comparable results to EC can be obtained. Currently, the IC is raising interest due to the increased number of high-quality SILs commercially available, even if they remain limited to the most classic compounds. To overcome this limitation, isotope standards can be generated in-house by derivatizing authentic analytes with labeled $^{13}\text{C}_2$ -dansylchloride and $^{13}\text{C}_2$ -dansylhydrazine.

As an illustration of the selection of quantification procedure, Figure 1.6 presents a flowchart applicable to LC-MS methods. Some parts of this flowchart are transferable to other methods of analysis and/or detection modes. Possible strategies are

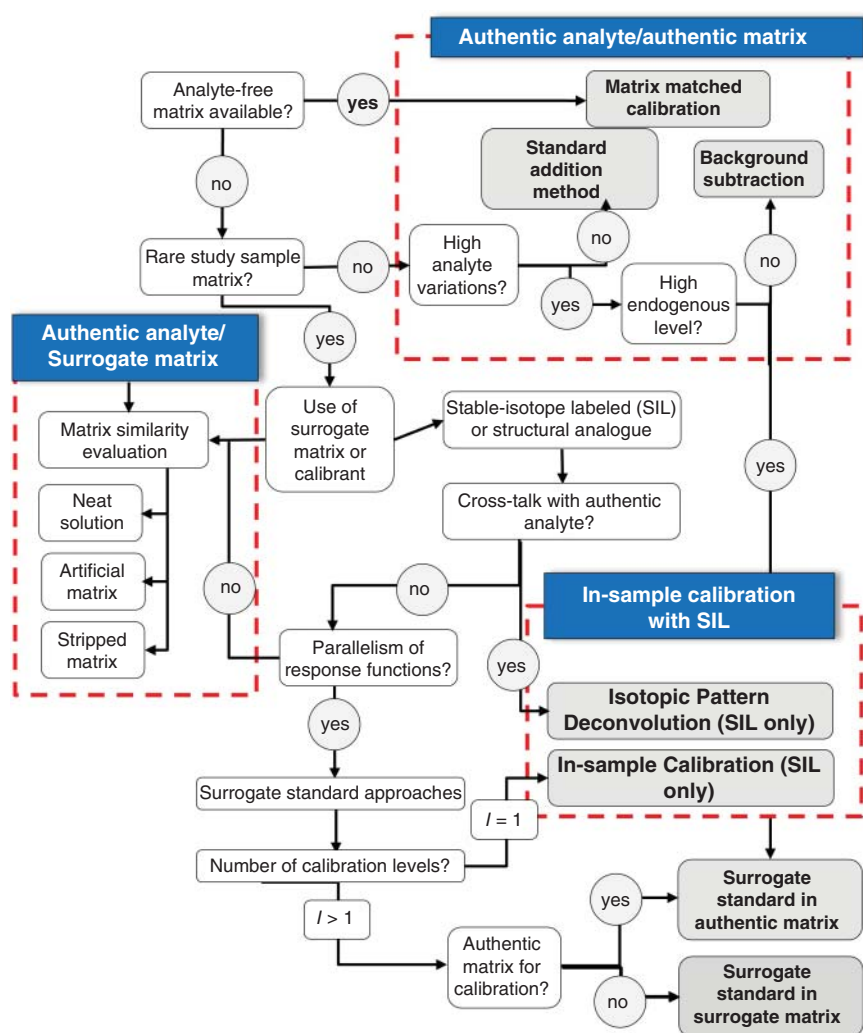


Figure 1.6 LC-MS on endogenous metabolites: proposed workflow for selecting a calibration operating procedure. Source: Adapted from Visconti et al. [13].

identified by square corner grey boxes, namely, authentic analyte/authentic matrix, authentic analyte/surrogate matrix, and ISC with SIL. For each case, different calibration procedures are appropriate, depending on complementary information about the analyte, the sample matrix availability, or the presence of endogenous analyte in the matrix.

More details are given in [13] and the rest of the chapter. In recent decades, advances in analytical calibration methodologies, instrument technology and enlarged SIL availability have contributed to improving the accuracy and throughput of quantitative analysis. However, the gap in knowledge between published official guidelines and strategies used by the analytical community prevents consensus about exactly how validation should be performed.

The introduction of innovative calibration approaches allowed the analyst to perform the calibration in the authentic working sample matrix, overcoming different bottlenecks such as the lack of blank matrices, the extraction efficiency, and matrix effect between the external calibration curve and unknown samples. Scientific interest is growing around direct internal calibration with SIL due to its analytical process simplicity and quickness to provide quantitative results from a few samples or even a single sample. With these unique advantages, internal calibration strategies have enormous potential to be widely applied for various quantitative applications and may even change the landscape of quantitative analysis, although these methodologies are still not officially endorsed by international guidelines for analytical method validation.

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